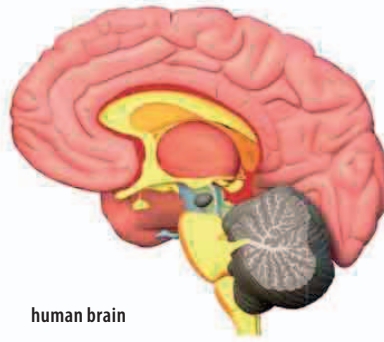


Physiology

Biologists are interested in the anatomy of living things—how bodies are made from tissues and organs. Physiology is the study of how an organism's anatomical features relate to a particular function. Physiologists may even study the fossils of extinct animals, such as dinosaurs, to make discoveries about their lives and deaths.



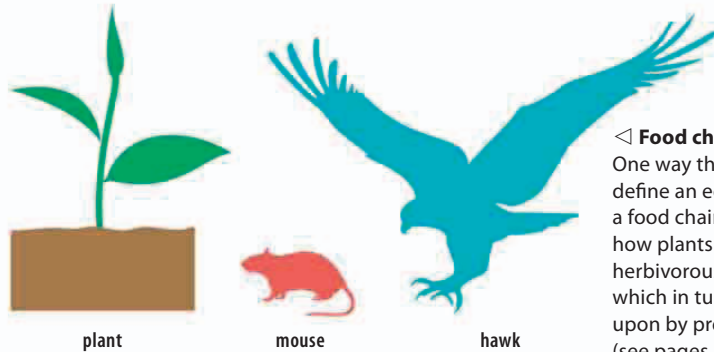
human brain

◁ Nerve center

The brain is a complex organ (a body part that has a specific function and is made of two or more kinds of tissue). The mass of nerve tissue is the main control center for the body (see page 68).

Ecology

The field of biology that investigates how communities of organisms live together is called ecology. Ecologists group wildlife into ecosystems, which occupy a specific living space or habitat. Scientists try to figure out the complex relationships between the members of an ecosystem. They may use their findings to help protect the habitat and its inhabitants from harmful human activities.



plant

mouse

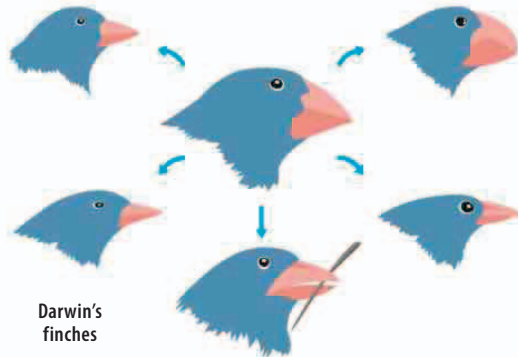
hawk

◁ Food chains

One way that ecologists define an ecosystem is by a food chain, which tracks how plants are eaten by herbivorous animals, which in turn are preyed upon by predators (see pages 76–77).

Evolution

Biologists have discovered that living things can change, or evolve, to adapt to new habitats. The process is very slow, but it explains why the fossils of extinct organisms share features with today's wildlife. Evolution also explains how similar animals such as these finches have become slightly different from each other in order to suit how they live.



Darwin's finches

◁ Bill shapes

These species of Darwin's finch each target specific types of food, such as seeds or insects. As a result, their bills have all evolved into different shapes (see page 82).

Conservation

The more biologists reveal about the natural world, the more they find that many species are under threat of extinction. While extinction is a normal part of evolution, it appears that human activities, such as farming and industry, are making species die out much faster than normal. Conservationists use their knowledge of biology to protect endangered species and prevent unique habitats from being destroyed.



giant panda

◁ Saving species

Without conservation, the giant panda, a bamboo-eating bear from China, may have become extinct. It was threatened by hunting and loss of its mountain habitat.